THE SECOND BRAIN FOR INVESTMENT BANKING

From Parametric M&A Discovery to Adaptive Agentic Graph Traversal

CENTRAL PROPOSITION

Strategic opportunity may reside in the relationships connecting a firm's capabilities, owners, licenses, markets, counterparties and dependencies. A disciplined Second Brain makes those paths inspectable. A parametric engine traverses them by predefined rules; an adaptive agent can choose investigative paths within independently enforced evidence and eligibility constraints.

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Second Brain Laboratory Series | Lab 03 | Pedagogical Research Monograph

All company identities, financial indicators, licensing records and benchmark outcomes are synthetic.

Abstract

This research monograph develops a synthetic M&A discovery environment for Aurora Financial Group. Its mandate is to investigate complementary cross-border payments, real-time settlement and digital-identity capabilities under explicit ownership and licensing constraints. The teaching architecture begins with reproducible company generation, builds an Obsidian-compatible vault, derives typed and task-specific knowledge graphs, conducts constrained traversal, compares four ranking policies and constructs structured packages for investment-committee judgment. It then evaluates validation and adverse-information handling before introducing a more autonomous alternative: LLM-planned adaptive graph exploration.

The primary technical source is the accompanying TOPIC 793 Colab notebook, including its eleven numbered code cells and additional methodological notes. This text-only version omits the supplied figures. The company universe and its benchmark labels are synthetic. Descriptive advantages and theoretical research opportunities should not be confused with evidence of real-world deal effectiveness. The monograph preserves the distinct roles of source data, graph operations, hypothesis formation, model judgment and human authority.

RESEARCH QUESTION

Can a source-traceable corporate knowledge graph reveal strategic M&A complementarities that are missed by conventional screening, and can governed LLM-directed exploration discover useful additional hypotheses beyond fixed parametric traversal?

TECHNICAL COMPANION

The authoritative technical record is the accompanying TOPIC 793 Lab 03 Colab notebook. This monograph follows its eleven code cells; the supplied cell figures are omitted from this text-only version. The notebook contains additional methodological discussion that can be explored independently.

Introduction: Acquisition opportunities as relational hypotheses

A conventional corporate screen organizes firms by size, geography, profitability and valuation. These are necessary descriptors, but they cannot alone disclose whether a target would unlock an existing distribution network, remedy a licensing bottleneck, create access to critical infrastructure or introduce a dependence on a rival-controlled provider. The Second Brain is designed to make such connections explicit, queryable and reviewable. Unlike a single long prompt, it keeps a durable store of entities and evidence, reconstructs relationships through typed edges and supplies narrowly selected context to model-based judgment.

The laboratory is deliberately synthetic, with a universe of two thousand companies and known benchmark opportunities. This enables transparent comparison of screening methods and reproducible demonstrations of adverse-information updates. The research objective is to teach the engineering of strategic discovery and its safeguards, rather than to claim predictive success in actual mergers and acquisitions. The eleventh cell changes the fundamental research question: instead of asking only what a predetermined algorithm finds, it asks how an LLM can propose, choose and revise graph investigations while remaining accountable to the same underlying evidence.

Cell 01 — Reproducible environment and institutional boundaries

The first implementation step establishes an explicit execution environment. Research notebooks often become difficult to audit because important settings are hidden in the analyst's session: random seeds, output directories, model identifiers, credentials, and package versions are left implicit. This laboratory instead creates a single root directory and a reproducible set of subdirectories for the synthetic universe, vault, graph outputs, reports, and evaluation results. The root is exactly the Google Drive location specified for TOPIC 793. When the notebook is run outside Colab, a local fallback path is used without pretending that Google Drive has been mounted.

The synthetic case will be generated deterministically from a fixed seed. That choice enables an analyst to re-run graph construction, inspect an individual firm note, and reproduce a candidate ranking without network access. The LLM remains the one deliberately non-deterministic component: API availability, provider model lifecycle, and generated prose may vary. To avoid misrepresenting this difference, the notebook records its requested model and whether a live call actually succeeded. We do not assume that a model identifier is valid merely because it has been requested. In particular, `claude-haiku-5-5` is used as provided and is checked transparently when the model call occurs. The notebook will never silently replace it with another model.

We install lightweight, standard Python packages for tables, graph operations, and textual ranking. The core data generator, vault writer, graph validator, search methods, and benchmark can therefore operate independently of Anthropic credentials. This separation is important institutionally: a temporary API outage should not destroy the ability to inspect the bank's factual records or reconstruct its analytical pipeline. Artifacts are written in conventional, inspectable formats—CSV, JSON, Markdown, and GraphML where practical—rather than requiring a proprietary graph database.

Finally, the cell prints the root path and creates an execution manifest. In a production research environment this manifest would also record source releases, ingestion timestamps, relevant corporate confidentiality classifications, and model-governance approvals. Here it captures the minimum needed for educational reproducibility. Later cells build on these same paths, so students should run this cell before any of the remaining nine. No market data are downloaded and no real companies are represented by the synthetic results.

A methodological boundary should be declared before any company enters the data set. A synthetic investment-banking laboratory can rigorously test software behavior, evidence provenance and the sensitivity of rankings without claiming that its imaginary firms resemble an observable takeover market. This distinction also protects the student against overinterpreting apparently precise scores. The reproducibility seed fixes the random draws used to create company records, but reproducibility of construction is not the same as realism of the underlying economics. A second execution with a different seed offers a useful sensitivity exercise: one should ask whether the qualitative contrast among retrieval architectures survives reasonable changes in the generated corporate universe.

The directory design should be understood as a miniature governance mechanism. Raw input tables belong in one location; the Obsidian vault, machine-readable graphs, evaluations and committee reports belong in separate destinations. This arrangement makes it possible to inspect which evidence supported a result without reverse-engineering the final narrative. It also distinguishes files that can be regenerated deterministically from material requiring model inference. A production implementation would require additional version control, access permissions and stable storage, but the educational principle is already visible: provenance is easier to preserve when artifacts are deliberately separated.

The execution environment also establishes a sensible boundary for external model calls. An API key belongs in the managed Colab secret store rather than a notebook output or hard-coded variable. Optional access to Claude should not be necessary for building the vault, validating graph structure or comparing deterministic candidate rankings. When the external model cannot be reached, the research pipeline should report the failure explicitly, retaining the evidence package for subsequent review. This is preferable to substituting a fabricated assessment or quietly altering the model identifier. Students should therefore learn to identify which outputs are deterministic, which are requested from an external service and which remain unavailable.

The conceptual lesson is that engineering context precedes exercising judgment. A strong language model cannot compensate for a missing relationship, an outdated license date or an undocumented shareholder assertion. Cell 01 is consequently more than notebook housekeeping. It defines a controlled research environment in which later assertions can be traced back to reproducible inputs, known computational transformations and clearly marked uncertainties.

Interpretation and evidence boundary

The graph, database and Markdown vault represent synthetic research inputs; a retrieved relationship is not itself proof of acquisition feasibility, economic value or regulatory approval. Read this stage in conjunction with the notebook execution and its recorded artifacts.

Cell 02 — Strategic mandate and synthetic corporate universe

An acquisition opportunity is meaningful only relative to a mandate. We therefore start by defining a fictional financial-services acquirer whose strategic ambitions span payments, market infrastructure, and adjacent technologies. Its gaps are represented as explicit target capabilities rather than buried in a paragraph of banker intuition. The generator then creates a heterogeneous universe of two thousand synthetic firms operating across twenty-five jurisdictions. Each firm has a sector, jurisdiction, scale, financial characteristics, descriptive text, capability profile, primary product, and high-level license position.

The universe is intentionally not a fully random spreadsheet. A realistic demonstration requires structured asymmetry: an operating company may hold a valued technological capability while a separate group company owns the license; distribution access may come from a partnership rather than legal ownership; and a target that appears excellent on headline metrics may be disqualified by control or regulatory restrictions. We plant a small set of benchmark opportunities so that retrieval quality can later be measured. We also plant deceptive targets whose apparent attractiveness conflicts with a hard acquisition constraint. These records are part of a controlled experiment, not hidden knowledge that the LLM is expected to invent.

Each corporate record is assigned a stable identifier and a human-readable fictitious name. Financial attributes are illustrative, internally bounded numbers and not estimated enterprise values. The universe includes regulated intermediaries, software businesses, banks, exchanges, infrastructure providers, and related companies, but the counts are pedagogical rather than empirical. The generator embeds enough semantic variation to support an alternative text-similarity baseline. It also creates structured fields that will later become nodes and edges. This dual representation is deliberate: the same facts must support conventional tables, text retrieval, and graph retrieval, otherwise the benchmark would compare incompatible information sets.

The cell exports a compact reference dataset to CSV and records the synthetic ground-truth IDs separately. This facilitates two modes of teaching. Students can examine a company profile in isolation, using conventional finance methods, and then inspect what the relationships reveal once the same profile is situated in an ecosystem. They should anticipate a core finding: the strongest financial filter is not necessarily the strongest acquisition thesis. Strategic relevance may depend on a subtle complementarity between capability, permission, and distribution that a company-level screen cannot represent adequately.

The strategic mandate gives a search engine a purpose, but it should not be confused with a complete theory of acquisition value. Aurora Financial Group is interested in cross-border payments, real-time settlement and digital identity, with five regions of particular interest. These specifications are enough to construct a controlled screening experiment. They are not enough to determine achievable synergies, negotiation success, transaction financing or the value of post-merger integration. The distinction will matter when the notebook moves from target identification to a committee memorandum.

The synthetic population deliberately combines conventional financial indicators with strategic and regulatory characteristics. Revenues are produced using a skewed positive distribution; growth, margins and valuation multiples follow bounded statistical draws. This supplies heterogeneous input data while keeping obviously pathological financial observations within pedagogical limits. Students should not read these distributions as calibrations to the real population of payments firms or bank holding companies. They simply provide a stable laboratory in which financial and network-based screens can be compared on the same observations.

The planted opportunity and decoy groups serve different evaluative functions. The opportunity firms are assigned relevant strategic capabilities and favorable financial features; the decoys are also financially attractive, but fail hard ownership or licensing constraints. By introducing both, the exercise prevents a model from winning solely by recognizing superficially attractive balance-sheet features. A meaningful discovery system should find opportunity while rejecting infeasibility. Yet an important caveat follows: if the planted opportunities directly match the signals favored by the graph algorithm, strong benchmark recall demonstrates alignment with the synthetic construction, not validated effectiveness on unseen real transactions.

One useful classroom variation would hold the financial columns fixed while rearranging capabilities and market positions. Another would preserve capabilities but alter control rights and the validity of licenses. These counterfactual constructions isolate what kinds of information drive the rankings. More ambitious extensions could introduce unequal information quality, shareholder coalitions, nonassignable agreements and economically meaningful dependencies. Cell 02 thus defines both the scope of the model and the limits of its inference. All later findings depend on what

kinds of strategic reality this initial synthetic universe makes possible.

Interpretation and evidence boundary

The graph, database and Markdown vault represent synthetic research inputs; a retrieved relationship is not itself proof of acquisition feasibility, economic value or regulatory approval. Read this stage in conjunction with the notebook execution and its recorded artifacts.

Cell 03 — Obsidian vault and typed corporate relationships

The vault is the prime knowledge substrate of the Second Brain. In this cell, synthetic corporate facts are converted into Markdown notes that can be opened directly in Obsidian or inspected with ordinary text tools. Company notes contain structured YAML-like front matter, a concise business description, and wiki-style references to related entities. Separate notes are created for owners, technology capabilities, licenses, markets, products, source records, and the acquisition mandate. This schema matters because it distinguishes the economic object represented by a note from the many documents that may refer to that object.

An important design decision is that Markdown links do not carry sufficient meaning by themselves. A link between two entities may indicate ownership, technical dependency, an operating license, distribution rights, or mere mention. Accordingly, we maintain a structured edge ledger with source IDs, relation predicates, effective dates, and confidence fields. The human-readable vault is complemented by machine-readable CSV files, preserving portability while enabling programmatic integrity checks. For this demonstration, the source notes are explicitly synthetic. No company record should be mistaken for a securities filing or legal opinion.

The cell makes material features of the corporate ecosystem discoverable through linked notes. A candidate may link to a shareholder, a technology, a regulated permission, a partner, and a market. Where appropriate, parent-company edges and commercial relationships are given direction: who controls whom, who distributes for whom, and who depends on whom. Direction is not cosmetic. Traversing from a target to a controller can reveal an unavailable control right, while traversing in the opposite direction expresses a different claim. The edge ledger will be treated as authoritative for graph construction, whereas Markdown is the human inspection interface.

One practical learning objective is to make the costs of knowledge engineering visible. A system that generates thousands of files can still fail if names are unstable, relationships point to nonexistent nodes, or dated evidence is silently reinterpreted. Later validation stages therefore check all edge endpoints. The vault is not presented as an oracle; it is a maintainable institutional memory with explicit design obligations. At the end of this cell students can navigate the actual artifacts under the specified Drive root and inspect the note for any shortlisted synthetic company.

The vault is the primary human-readable memory substrate, while the edges ledger supplies machine-operable semantics. A Markdown file is not a complete graph database merely because it contains wikilinks. The relations have to be extracted, typed, checked and assigned identifiers. This distinction answers a recurring pedagogical question: why not simply let the language model read all company notes? A broad reading may produce fluent summaries, but it does not automatically yield consistent control relationships, valid effective dates, or an auditable path between a strategic need and a candidate. In the vault, a person can inspect one firm's story; in the graph, an algorithm can systematically examine a thousand comparable connections.

Cell 03 writes company, owner, license, source, capability, market and mandate notes. It also constructs typed links such as controls, possesses capability, serves market, holds license, distributes for and depends on. These predicates should be kept separate. An ownership edge is not equivalent to a supplier relationship, and neither proves that regulatory change-of-control permission is available. By preserving relation type and documentary source, the architecture supports specific follow-up questions rather than an undifferentiated network of proximity.

A provenance record identifies the synthetic source for an asserted edge; it does not establish real-world truth. An equally important distinction concerns the synthetic owner objects: the constructed 75 percent ownership edge is a teaching representation, not an actual beneficial-ownership determination. Likewise, the valid flag on an operating-permission node is merely a test condition. A real acquisition would require verification against current official registries, regulatory rules and transaction agreements. The notebook's pedagogical strength is that these different statements are at least explicitly represented rather than silently inferred.

Students should inspect the generated company notes alongside the edges table, comparing one written wikilink with its corresponding typed relationship. They should then perturb a record and ask whether the change is reflected consistently across the vault and graph. That experiment illustrates a general risk of duplicated knowledge representations: if one view changes and the other remains stale, the Second Brain can contradict itself. Production architecture should therefore designate an authoritative fact store and reproducible transformations. The educational vault demonstrates how an intuitive personal-knowledge interface can coexist with disciplined computational retrieval, without treating visual connections as a substitute for validated relationship semantics.

Interpretation and evidence boundary

The graph, database and Markdown vault represent synthetic research inputs; a retrieved relationship is not itself proof of acquisition feasibility, economic value or regulatory approval. Read this stage in conjunction with the notebook execution and its recorded artifacts.

Cell 04 — Validation and multiplex graph projections

The master graph is useful only if its ontology is precise. We load the typed edge ledger into a directed graph and confirm that every referenced node exists. Basic checks also cover duplicate identities, invalid ownership percentages, missing source identifiers, and unfamiliar relation types. We then derive several overlapping projections: ownership/control, capability and technology, regulatory permissions, commercial distribution, and transaction dependencies. Rather than collapse all edges into an undifferentiated network, these projections preserve the reason two entities are connected. A license edge is not economically equivalent to a supplier edge, and neither proves acquisition feasibility.

The derived graphs represent different views of one corporate reality. For example, a software target might be strategically desirable because its technology completes the acquirer's product portfolio. Its commercial graph might show extensive market access through partner relationships. Its ownership graph, however, may reveal that a controlling shareholder cannot be displaced. The regulatory graph may require a transfer approval, and the technical graph may show concentration on a proprietary platform. These dimensions should not be averaged away into a single network centrality measure.

The notebook provides node and edge counts for each projection and exports a compact graph structure suitable for further exploration. Every edge retains the source note ID and the time or status known to the synthetic generator. Such provenance is foundational to responsible retrieval: the analyst should always be able to ask why the graph claims a relationship and where it originated. Here, each source is synthetic, but the audit pattern is the same one a bank would apply to genuine disclosures, verified contracts, permissions, and internal reports.

This cell also demonstrates the architectural distinction between Obsidian and a graph database. The Obsidian vault makes knowledge legible, portable, and editable. The parsed graph makes it computationally traversable. The validation rules make it auditable. A language model is not needed to carry out any of these operations, and introducing one prematurely could make deterministic quality checks less reliable. At this stage, students should be able to inspect the topology, understand the ontology, and identify how a company can occupy very different positions in different relational layers.

Cell 04 creates a deliberately explicit bridge between data integrity and analysis. The first checks protect against duplicate company identifiers, missing endpoint nodes, unknown evidence references and nonsensical numeric fields. These tests do not prove that the underlying propositions are economically correct. Instead, they establish that the constructed network is internally interpretable. A graph with broken endpoints or implausible ownership shares cannot support an investment investigation, however persuasive its later narrative may appear.

The master structure is a directed multigraph rather than a simple undirected network. Direction matters because an owner controls a company, while the company does not thereby control the owner. Multiplicity matters because the same pair of entities may carry multiple different relationships. Preserving edge attributes also allows the analyst to distinguish a relationship's type, source, confidence and stated validity date. These choices create the foundation for more sophisticated temporal and evidence-quality controls, though the laboratory does not implement a complete historical corporate-event database.

The five projections answer different investigative questions. The ownership view isolates control; the capability view relates strategic needs to technological attributes; the regulatory view organizes licenses and markets; the commercial view concentrates distribution and dependency relationships; and the diligence view combines relations relevant to transaction feasibility. Every projection retains the master graph's node set while selecting its own subset of edges. Consequently, equal node counts across views do not imply that every node participates in a meaningful connection in each view. Students should measure active nodes separately if they want to compare actual coverage.

A practical classroom exercise is to select a target with a commercially attractive profile and inspect how that same target appears across all five views. Its capability neighborhood may be rich while its ownership view is simple. Alternatively, a licensed company may depend on a strategically critical counterparty. Those contrasting views expose the difference between describing an institution and deciding whether it is acquirable. A relational database can store the same facts and compute equivalent joins; the graph's advantage here is expressive traversal and visual inspection, not the assertion that SQL is incapable of modeling relationships. The correct architectural choice depends on query workload, maintenance cost, explainability and organizational governance.

Interpretation and evidence boundary

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Cell 05 — Constrained traversal from gaps to acquisition hypotheses

Traversal is the analytical core of Lab 03. Instead of searching the graph from an arbitrary node and reporting its neighbors, we start from a mandate-defined missing capability. This produces a directed and economically meaningful question: which companies supply the capability, through which ownership and licensing arrangements, with what distribution opportunities, and subject to which structural impediments? The traversal engine returns bounded paths and human-readable explanations, not merely node IDs or a centrality statistic.

We deliberately limit the types of edges and the maximum number of steps. Unconstrained graph walks quickly lead to irrelevant, misleading associations—two companies might share a jurisdiction, for instance, without possessing any strategic complementarity. The path grammar privileges capability ownership, licensed activity, commercial distribution, and control dependencies. It records all intermediate nodes so that an analyst can evaluate whether the inferred chain is a plausible acquisition mechanism or only a coincidental association. For our synthetic example, the principal path begins with the client's capability gap, proceeds to a technology or product node, identifies companies holding that capability, and then extends to distribution partners, licensed markets, shareholders, and technical dependencies.

The system treats the path as a hypothesis rather than as proof of transaction value. A valid path can establish that the target owns a technology and holds a relevant permission according to the synthetic ledger. It cannot establish future integration success, antitrust clearance, bid price, or transaction certainty. Those conclusions require additional commercial, legal, and financial analysis. We explicitly distinguish evidence paths, constraints, and unknowns. Traversal is designed to make adverse findings visible, including firms whose recorded shareholder or license state makes acquisition infeasible even if the strategic bridge looks attractive.

The cell displays selected traversal paths for planted benchmark candidates and measures their lengths. This is the moment students should compare a graph-based argument with a standard company screener: the screener says what a company is; the traversal says why a combination might matter. The difference is especially important in cross-border financial services, where permissions, ownership, technology dependencies, and market access often determine the strategic value of a transaction more than a static industry label does.

This cell introduces the most important distinction in the deterministic laboratory: not every displayed step of a strategic path is an observed graph edge. Aurora's need for digital identity is linked to a candidate's possessed capability by a deliberate matching operation. The transition from a gap node to a capability node is analytical logic rather than a corporate fact stored in the vault. Treating the transition as though it were independently documented would create a misleading chain of evidence. By labeling it explicitly, the notebook separates the rules of discovery from the evidence discovered.

The traversal starts with the acquirer's predefined needs and looks for firms whose primary or secondary capabilities correspond to those needs. For qualifying candidates, it gathers selected direct relationships: licenses, served markets, distribution partners and dependencies. The output includes readable path descriptions and an out-degree indicator. These objects help a student explain why a candidate entered the search results and identify facts requiring verification. They are not a general shortest-path optimization nor a proof of contractual or economic connectivity between arbitrary nodes.

An illustrative capability chain has four displayed node positions: acquirer, strategic gap, capability and company. It is more accurate to describe this as a four-node discovery explanation with a derived matching step than as four observed network hops. The direct edges collected thereafter enrich the thesis but do not independently demonstrate that the acquisition would deliver the hoped-for customer access or technological interoperability. A commercial relationship could be exclusive, revocable or economically immaterial. Likewise, a license could be valid today but require discretionary approval following a proposed transfer of control.

The pedagogical limitation motivates Cell 11. Because the initial traversal rules are fixed, the system is strongest when attractive firms directly match the predetermined capability vocabulary. It is weaker at investigating a hypothesis in which value resides in an intermediary, a coalition of complementary firms or a market-access bottleneck. The original graph might already contain the relevant relationships, yet its search policy could fail to explore them. Students should distinguish coverage of the knowledge base from coverage of the hypothesis space: a complete graph does not guarantee an imaginative traversal. This insight leads naturally from parametric screening toward governed agentic exploration.

Interpretation and evidence boundary

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Cell 06 — Four discovery engines and comparative evaluation

The lesson would be incomplete if it simply asserted that graph retrieval is superior. This cell creates a controlled competition among four discovery procedures operating over the same synthetic corporate facts. The conventional screen combines simple financial and sector criteria. The semantic baseline uses TF-IDF similarity between the strategic mandate and company business descriptions. The graph-guided method evaluates directly relevant capability holdings and the supporting paths through ownership, permissions, and distribution. A hybrid model blends the semantic signal with constrained relational strength and strategic complementarity.

The benchmark contains planted positive opportunities and deliberately deceptive candidates. Its purpose is to show how alternative retrieval architectures can behave under the same data-generating assumptions. We report precision at a specified shortlist size and recall of the planted opportunities, together with any constraint violations in the selected list. These measures are meaningful only within the synthetic benchmark. We do not infer real-world improvements in analyst productivity, return on capital, M&A completion rates, or valuation from fabricated records. Indeed, one should expect the graph method to benefit from the way the benchmark was designed; that is a hypothesis for future testing with independent data, not a general empirical finding.

A key distinction is between candidate retrieval and transaction selection. Retrieving a valuable strategic bridge does not mean the acquisition is feasible. The ranking stage therefore exposes eligibility, so that candidates failing hard rules cannot enter the viable investment-committee shortlist. In practical bank governance, a very high expected synergy would not compensate for a legal prohibition or structurally unavailable control. The screen also guards against interpreting raw node degree as a strategic score. Connectivity creates possibilities, but it can also indicate operational concentration and negotiation complexity.

For teaching, this cell makes methodological assumptions transparent. The tabular weights are documented and deliberately basic. Text ranking measures descriptive affinity, not business value. The graph rank uses validated relations, not unexamined LLM claims. The hybrid combines complementary signals under an eligibility gate. Students can vary the benchmark seed, weights, shortlist size, or planted opportunity structure and observe how the conclusions change. That sensitivity analysis is more valuable than selecting a single method as universally optimal without evidence.

The comparison among four methods is a controlled test of retrieval policy, not a competition among general-purpose artificial intelligences. The tabular screen combines normalized growth, margin and valuation factors. Semantic retrieval applies TF–IDF to synthetic company descriptions and evaluates cosine similarity to a mandate-derived query. The graph signal rewards matching strategic gaps and selected network relationships. Finally, the hybrid score combines normalized graph and semantic information using explicit weights. Because all methods share the same firm universe and hard eligibility gate, the rankings expose differences caused by how information is prioritized.

The validity of an experiment depends on what its metrics mean. Precision at twenty measures the number of planted positive firms in a twenty-name shortlist divided by twenty. Recall at twenty divides that same hit count by the five benchmark opportunities. These measures are valuable for comparing methods in this synthetic environment, but they are not external deal-origination success rates. The planted companies were selected and modified before evaluation, so the model can be favored by the design of the test. This limitation should be stated alongside every displayed recall percentage.

The eligibility gate is equally important. Companies without available control rights or valid license indicators receive a disqualifying score for purposes of ranking. The gate is applied identically to all four methods. Thus the comparison does not inadvertently reward a retrieval mechanism for finding impressive but explicitly infeasible decoys. However, these Boolean indicators greatly simplify actual acquisition feasibility. A genuine transaction requires examining ownership chains, shareholder agreements, approvals, economic restrictions and financing conditions, often with uncertain outcomes.

Students should investigate the sensitivity of rankings to normalization and blending weights. A modest change in the graph weight could alter the order of firms with close scores. Similarly, substituting richer embeddings for TF–IDF could affect semantic ranking without changing the corporate facts. The graph approach should not be declared universally superior on the basis of one scoring recipe. Its educational value is that it explicitly represents which evidence channels are being rewarded. The most robust finding is methodological: different search policies expose different dimensions of the same knowledge base, and no score should be interpreted without the assumptions that created it.

Interpretation and evidence boundary

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Cell 07 — Evidence packages and investment committee context

A second brain should not merely return names; it should assemble a decision-ready context. In this cell, each shortlisted target receives a structured evidence package containing its company profile, relevant graph edges, path-based strategic thesis, control status, license conditions, commercial opportunities, risks, and outstanding diligence questions. The package also names the source notes supporting material claims. This is a fundamental shift from open-ended prompt engineering to engineered, inspectable context selection.

The package is intentionally compact. Feeding every company note and every graph edge into a language model would make the system costly, slow, and susceptible to inconsistent attention. More important, indiscriminate context can obscure decisive adverse evidence. Instead, the assembler prioritizes information directly relevant to the transaction thesis and its feasibility. For each candidate it includes positive reasons for consideration and potentially disqualifying factors. A valid permission is not enough if regulatory consent to a change of control is unresolved. A strong capability match is not enough if the essential technology depends on a rival-controlled infrastructure provider. A promising shareholder structure is not enough if minority protections could block the deal.

The evidence pack is serialized to JSON, making it reproducible and reviewable before any LLM call. It is also summarized in a human-friendly format for students. A rigorous M&A process would add valuation models, tax considerations, contractual assignment conditions, sanctions screening, jurisdiction-specific counsel, and management diligence. Our teaching system instead demonstrates the discipline of identifying which facts are available, what inferences they support, and which questions remain open. It does not claim that the synthetic records amount to a complete legal or financial diligence data room.

An important governance property is that evidence packages can survive model replacement. If Claude is updated or a bank uses another provider, the graph paths, IDs, and source records can remain stable. This separates the institution's knowledge substrate from its current inference engine. The model is asked to use the package, not to invent missing counterparties or conclude that a regulatory approval has been obtained. By the end of the cell, a student can point from a proposed deal thesis back through a path to the original Markdown note that supports its constituent relationships.

The evidence package is the institutional boundary between search and judgment. A shortlist alone communicates very little: it does not say why each target appeared, which permissions matter, what dependencies were encountered or which documents can substantiate the recommendation. Cell 07 collects these ingredients into machine-readable dossiers for three selected companies. The package includes a financial profile, matched strategic gaps, licensing and control flags, selected typed edges, illustrative paths, admissible source identifiers and diligence questions.

A distinctive feature is that the package retains both positive and potentially adverse information. The existence of a matched capability may support a transaction thesis, while a technological dependency or change-of-control review requirement may weaken its feasibility. Strong investment-banking analysis must preserve those tensions. The architecture should resist the temptation to assemble only supportive observations for the LLM. A system that retrieves evidence selectively to justify a favored candidate would be efficient at producing narratives but poor at making decisions.

The package is deliberately bounded. It includes only a subset of a company's relevant direct edges, not every possible relationship in the vault. This helps manage context size and costs, but introduces a selection risk. If a critical adverse relationship falls outside the retained subset, the model cannot account for it. A production diligence process would use relation-specific priorities, adverse-evidence search and coverage checks rather than relying on the first fixed number of encountered edges. The analyst should record not only included evidence but why other facts were excluded.

The data also contain two forms of constraint that must remain separate. Control availability concerns whether the hypothetical ownership position is obtainable; license validity concerns an existing operating permission. Neither is equivalent to approval of an acquisition or the certainty of synergy realization. The diligence prompts therefore request confirmation of beneficial control, regulatory approval requirements, technology integration, supplier concentration and valuation assumptions. Cell 07 does not answer those questions. It ensures that they are passed explicitly to the next stage. That is how a Second Brain can improve the institutional quality of AI judgment without turning retrieval into an unsupported recommendation.

Interpretation and evidence boundary

The graph, database and Markdown vault represent synthetic research inputs; a retrieved relationship is not itself proof of acquisition feasibility, economic value or regulatory approval. Read this stage in conjunction with the notebook execution and its recorded artifacts.

Cell 08 — Governed Claude judgment and structured reporting

The language model enters only after the retrieval and context-assembly stages have completed. Its function is not to rebuild the corporate database from memory or speculate freely about the fictional companies. Instead, it receives a bounded, serialized package for three viable candidates and a disciplined investment-committee task. The desired output is a structured memorandum containing candidate assessments, transaction theses, possible deal structures, principal synergies, execution risks, diligence requests, reasons for rejecting alternatives, and an explicit uncertainty statement. The prompts prohibit unsupported statements of fact and require citation to the synthetic source IDs already provided.

The requested model for this notebook is `claude-haiku-5-5`. Because vendor identifiers and access entitlements can change, the code exposes the model in one configuration variable and reports API errors without automatically substituting a different one. In Colab, the Anthropic key is read from Secrets under `ANTHROPIC_API_KEY`. There is no embedded key in the notebook. The Anthropic SDK call avoids the `temperature` argument entirely, respecting prior experience with provider incompatibilities, and includes a deliberately finite output-token budget. Responses are parsed defensively: the code attempts strict JSON parsing, handles optional Markdown fences, and reports failures rather than treating malformed output as validated judgment.

The model may be unavailable even when the deterministic graph experiment succeeds. In that case the notebook still generates the evidence package, a template committee analysis, and a clear status label: no live LLM judgment was produced. This distinction prevents a particularly serious teaching error—presenting heuristic text or deterministic diagnostics as though they were actual Claude conclusions. The model invocation uses a direct, single request; no autonomous recursive calls are hidden in this stage.

Once a live response is available, the notebook stores the raw text, the parsed structure if possible, and model metadata in the outputs directory. The result is a first draft for analyst review, not an investment approval. A regulated financial institution would normally impose additional internal controls on model data flows, confidentiality, model risk, supervisory permissions, conflicts, and committee sign-off. These constraints are outside the synthetic laboratory but frame the proper interpretation of its output.

The language model is introduced only after the knowledge system has performed its retrieval and packaging work. This is a deliberate allocation of responsibilities. Claude does not design the fixed screening rules, create the synthetic ownership graph or certify its own sources. Its role is to interpret bounded company-specific evidence in a form that resembles the agenda of an investment committee. The requested fields include transaction thesis, possible structure, synergies, risks, diligence requests and source identifiers, each with compact output constraints. The corrected notebook processes the selected companies individually rather than demanding an oversized combined answer. This change responds to a common operational failure: a model can begin producing syntactically correct JSON and still be interrupted before closing the object when the output budget is exhausted. The stop reason must therefore be inspected before parsing is considered successful. Independent calls reduce the likelihood that one long memorandum consumes the entire allowance and conceal which candidate caused the error. Per-candidate usage and raw responses are retained for subsequent diagnosis. A structured JSON schema does not guarantee factual correctness. The model may cite an allowed source identifier while drawing a conclusion the source does not support. It may discuss feasible acquisition structures without enough contractual information to establish viability. Therefore, output parsing, source-admissibility checks and substantive diligence should be regarded as separate tasks. Automated validation can reject unknown identifiers and incomplete objects; it cannot alone certify legal or commercial truth. The model configuration itself is also an experimental variable. The requested Claude Haiku identifier must be accepted by the provider in the actual execution environment, and token limits must be compatible with the model's current capabilities. If no API key is available, or if a response terminates unsuccessfully, the correct result is an explicit absence of live model judgment. Generating a fallback investment opinion and labeling it as Claude's result would corrupt the audit trail. The broader institutional principle is straightforward: high-quality decision systems must be able to report uncertainty and failure without losing the valuable deterministic context already assembled.

The notebook also makes an operational distinction between a model response that is incomplete and a model response that is unsuitable for the investment committee. A parseable object can still be inadequately supported; a truncated response may contain promising ideas without constituting an auditable conclusion. Neither should be silently promoted to approved judgment.

Interpretation and evidence boundary

The graph, database and Markdown vault represent synthetic research inputs; a retrieved relationship is not itself proof of acquisition feasibility, economic value or regulatory approval. Read this stage in conjunction with the notebook execution and its recorded artifacts.

Cell 09 — Validation, hard constraints, and adverse-information stress

A fluent committee memorandum is not necessarily a supported one. This cell formalizes an important separation: the LLM proposes a judgment, while deterministic checks test the structure and evidentiary status of its claims. The validator verifies the expected JSON fields, candidate IDs, and cited source-note identifiers. It checks that the selected candidates remain eligible under the non-compensable control and licensing conditions. It also flags any source reference not present in the provided evidence package. These tests do not certify legal correctness or strategic wisdom; they are mechanical safeguards intended to catch frequent failure modes.

The strongest governance principle is that a candidate must not be promoted simply because the model wrote a persuasive narrative. If the graph records unavailable control rights, the hard constraint remains operative. If the model refers to a missing permission or nonexistent source record, the notebook identifies it as unsupported. If no live LLM request succeeded, validation clearly says so. It neither awards a fictitious pass nor infers a judgment from deterministic rankings. This behavior is particularly important after previous notebook experiences in which an API error could otherwise leave a superficially plausible success message on screen.

The cell also runs a counterfactual stress test. One shortlisted target is assigned a newly discovered adverse dependency, simulating an updated data-room document. The relevant decision flag is changed and the selection engine is rerun. This demonstrates that knowledge updates can alter the ranking and admissibility of a transaction without requiring the analyst to rebuild all company profiles or re-query the entire universe. In a live system, the change would require a dated source, review workflow, and graph-version history. Here it is an explicit synthetic test with a separately saved result.

Finally, the validator writes an audit file that distinguishes deterministic findings from model-generated content. The critical learning outcome is not that automation can validate its own conclusions. Rather, it is that independent controls can prevent some predictable errors while preserving the need for legal counsel, valuation specialists, due-diligence teams, and investment-committee judgment. A second brain improves the structure of inquiry; it does not replace accountable decision-makers.

Validation in Cell 09 has two complementary components: checks on the produced assessment and a counterfactual change to candidate eligibility. The first verifies that the LLM output, if present, contains recognizable assessment records for known shortlisted companies and does not cite source identifiers outside the admissible sets. The second tests whether the deterministic selection procedure responds appropriately when the leading candidate becomes ineligible. Together they make a modest but useful bridge between static reporting and a process capable of absorbing adverse information.

The logic is not comprehensive semantic verification. A citation identifier may belong to the supplied package without proving that a particular sentence follows from the underlying source. Nor does a passed source check mean that all necessary documents were retrieved. The code should therefore be interpreted as a mechanical assurance layer, with expert review still required to examine ownership evidence, regulatory implications, contractual exceptions and valuation assumptions. The distinction between syntactic validity, evidence admissibility and substantive correctness is central to the training of financial practitioners.

The adverse-fact test is deliberately sharp: control availability for the leading target is switched to false while the underlying scoring method remains unchanged. The system then recalculates eligibility and identifies a revised shortlist. This isolates the effect of a binding constraint from the effect of financial attractiveness. A company can retain its high capability score and still be properly rejected. In real negotiations, new information might be more ambiguous. An unexpected voting agreement, a conditional approval or a financing restriction may not translate neatly into a Boolean field. Nonetheless, the simplified test demonstrates the right architectural instinct: new adverse evidence should be allowed to defeat an appealing thesis.

Students should compare the original and revised results at three levels. At the data level, identify precisely which field changed. At the model level, determine whether the shortlist change follows from the eligibility rule rather than an unexplained model decision. At the governance level, ask which additional diligence would be required before a human committee endorsed the new ranking. A Second Brain is valuable not merely when it remembers facts but when it makes the consequences of factual changes traceable and contestable.

Interpretation and evidence boundary

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Cell 10 — Final laboratory report and institutional interpretation

The final cell consolidates the laboratory's artifacts and produces a concise, reproducible report. The output directory contains the synthetic company universe, the Obsidian-compatible vault, the typed edge ledger, graph summaries, candidate rankings, path evidence, investment-committee evidence packs, available Claude responses, validation results, and stress-test updates. Together, these files show the entire transformation from an ambiguous strategic mandate into an evidence-backed shortlist. The final report points to the same root directory, making it straightforward to inspect the outputs later in Google Drive or load the Markdown directory in Obsidian.

The benchmark summary is framed carefully. Retrieval metrics are computed over planted synthetic ground truth. They are diagnostic of whether the experiment's algorithms recover the opportunities placed in the data, not evidence that any method has already improved real-world M&A decisions. The report explains how each screening paradigm performed and identifies the hard constraints that filtered apparently appealing companies. It also distinguishes graph completeness from inference quality: even perfect path reconstruction cannot establish synergy realization, the likelihood of regulatory authorization, deal price, or management compatibility. Those judgments need external verification and further analysis.

One can read the laboratory through a five-layer institutional architecture. Knowledge acquisition builds the factual inventory. Semantic representation and graph construction make facts connectable. Constrained traversal turns a mandate into plausible hypotheses. LLM synthesis expresses the alternatives for decision-makers. Validation, human review, and governance determine what can responsibly advance. These are separable modules: changes to the model need not invalidate the vault, and a newly learned ownership condition need not require redesigning the entire pipeline.

From an investment-banking perspective, the system is best understood as a structured origination assistant rather than a deal-execution robot. It can widen the search space, expose otherwise invisible strategic bridges, accelerate early diligence questions, and produce explainable committee materials. It cannot independently negotiate, certify title to shares, estimate binding synergies, or deliver legal opinions. Its most important intellectual contribution is to reframe a transaction as an interaction between complementary networks, subject to legal and economic constraints.

The final section also makes the notebook suitable for pedagogical reuse. Students can change the mandate, reassign capability ownership, withdraw a license, create a competing claimant, or discover a hidden dependency. They can then observe which graph paths disappear and whether the shortlist adapts. The experiment closes where a responsible second brain should: not with a declaration that the answer is certain, but with a traceable set of supported possibilities, rejected alternatives, missing information, and concrete next steps for expert judgment.

The final report is a consolidation artifact, not an independent source of authority. It brings together the strategic mandate, the size of the synthetic knowledge system, the comparative screening metrics, the principal candidate hypotheses, the model-judgment status and the adverse-information test. Its value arises from preserving an organized chain between evidence, computation and interpretation. A polished report with no accessible intermediate artifacts would not provide the same assurance to a reviewer or instructor.

The exported files permit several different forms of audit. Company and edges tables permit inspection of underlying data. The Markdown vault allows a reader to open an entity dossier and follow human-readable connections. Graph summaries and relationship ledgers expose the computational projections. Ranking tables and benchmark metrics make comparisons replicable. Evidence packages identify the bounded context shown to the LLM. Model diagnostics and validation reports document whether an actual model judgment was obtained and which automated tests were passed. These files should be archived with sufficient version information to reconstruct the notebook execution.

An investment committee should interpret the shortlist as a set of hypotheses. COMPANY_0017, COMPANY_0874 and COMPANY_0369 are synthetic teaching examples, not real targets or offers. Their financial attributes and licensing indicators do not supply an enterprise valuation, a financing plan, legal clearance or a negotiated transaction structure. Those stages require source documents, professional diligence, economic modeling and appropriately authorized governance. This qualification is not an embarrassment to the architecture; it is the correct institutional division of labor.

The deepest teaching result is that the Second Brain changes the object of search. Conventional screening asks which companies satisfy a set of attributes. Graph-oriented discovery asks which combinations of capabilities, permissions, owners, markets and dependencies may have strategic significance for a particular acquirer. The report is strongest

when its explanations preserve that relational argument. It should also state that a relational database could represent and query the same underlying facts. The value of the graph is not metaphysical superiority but practical clarity in multi-relation exploration, evidence tracing and the assembly of decision-specific context.

Interpretation and evidence boundary

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Cell 11 — Adaptive agentic discovery: LLM-directed traversal

An important distinction in this laboratory concerns the division of responsibilities between the knowledge graph, the traversal algorithm, and the large language model (LLM). Our implementation adopts a **predominantly deterministic, parametric architecture**. The corporate universe, relationship types, graph views, candidate-matching criteria, scoring weights, and eligibility constraints are specified in Python before the system begins searching for acquisition opportunities. The algorithms then execute those predefined procedures. The LLM enters subsequently, receiving evidence packages for shortlisted candidates and preparing an investment-committee assessment. Consequently, the LLM does not independently decide which relationships to explore, which graph paths to prioritize, or when the search has exhausted the relevant strategic possibilities. This separation is intentional. It provides reproducibility, computational efficiency, comparability across screening methods, and an auditable distinction between retrieving evidence and interpreting it. In a pedagogical setting, students can see precisely how the traversal operates before introducing the additional uncertainty of autonomous model-directed exploration. However, this design also imposes an important limitation: **the system can discover only those classes of opportunity that its predefined search logic is capable of recognizing**. For example, our algorithm starts with three strategic gaps: cross-border payments, real-time settlement, and digital identity. It searches companies possessing the corresponding capabilities and explores selected relationships involving markets, licenses, distribution agreements, and dependencies. But suppose an attractive acquisition opportunity could arise through a less obvious strategic combination: acquiring an insurance technology provider to obtain a distribution relationship with several banks, thereby gaining indirect access to payment customers. Such a hypothesis might require traversing relationship types or pursuing an intermediate strategic objective not anticipated by our original algorithm. The knowledge may already exist in the vault, yet the parametric traversal could fail to discover it. Yes. A more advanced architecture would allow the LLM to participate not only in evaluating acquisition candidates but also in **planning, directing, and revising the search through the knowledge graph**. We can distinguish three progressively more autonomous architectures. **Architecture A — Parametric traversal (our current laboratory).** The analyst specifies the strategic objectives. Python executes predefined searches and computes scores. The LLM receives a bounded evidence package and produces a judgment. Strategic mandate → Fixed traversal → Candidate ranking → Evidence package → LLM judgment. **Architecture B — LLM-planned traversal.** The analyst specifies the strategic mandate, but the LLM decomposes the problem into research questions and proposes graph operations. A

Cell 11 shifts the experimental variable from retrieval signals to control over the search policy. In the earlier cells, Python applies fixed matching criteria, traversal types, score weights and eligibility constraints. The language model then interprets a packaged shortlist. Under the new architecture, Claude receives the mandate and an approved set of graph operations, formulates a research hypothesis and chooses which tool to invoke next. The graph engine executes the requested action and returns factual observations. Claude evaluates those observations and determines whether to search, inspect, expand, compare or finish. This autonomy is bounded rather than unrestricted. The agent cannot execute arbitrary Python, create its preferred corporate facts or change the legal eligibility flags. Tool arguments and permitted relationship types are constrained, observations are limited in size, and every iteration is logged. These features allow students to identify whether a candidate emerged from actual evidence or from an unsupported model story. The investigator may design a question, but the repository controls the answerable facts. This separation is the crucial governance distinction between intelligent exploration and unchecked generation. An adaptive agent may explore indirect opportunities that the initial fixed score overlooks. For instance, a target with modest direct capabilities might occupy an unusually valuable position in a distribution network, or a superficially attractive technology provider might depend on a rival-controlled infrastructure component. These are possible search hypotheses, not verified findings of this notebook. Their discovery would require that the corresponding evidence exists in the synthetic graph and that the agent actually requests the relevant operations. The present environment also cannot establish transaction economics merely by observing network proximity. The comparison with the deterministic system should therefore be designed as an experiment rather than a promotional claim. Preserve the same corporate facts, opportunity labels, hard gates and evaluation questions. Run both architectures repeatedly; measure correct opportunity recovery, source traceability, false leads, number of graph operations, model token expenditure, runtime and stability. Critically, the agent may not be favored simply for searching more records: additional investigation must create incremental

Interpretation and evidence boundary

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Conclusion: From fixed traversal to governed strategic imagination

The comparison between parametric and agentic discovery should not be framed as a victory for autonomy. The deterministic architecture provides consistent screening, transparent scoring, fixed hard constraints and reproducible baselines. The adaptive architecture proposes and revises investigative operations using model-directed hypotheses, while the graph engine restricts execution to existing evidence. Both depend on the same synthetic facts, and both require transaction diligence and committee review. In a financial institution, a productive division of labor is plausible: broad deterministic screening followed by bounded agentic exploration of ambiguous, potentially consequential or strategically unfamiliar relationships.

A rigorous next experiment should preserve the exact same information set and benchmark labels, repeat agentic runs, and measure recall, evidence accuracy, false leads, latency, tokens, search breadth and the incremental value of new hypotheses. The notebook establishes a teaching architecture for such a comparison; it does not yet establish externally validated superiority of either approach. The fundamental lesson is institutional: knowledge representation determines what can be searched, traversal policy determines what will be discovered, and governed human judgment determines what can responsibly be recommended.

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